Knowledge, Practices, and Challenges of Infection Prevention and Control Among Nurses in Tertiary Healthcare Settings

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ABSTRACT

Background: Hospital-acquired infections remain a significant challenge in healthcare settings worldwide. Nurses, being at the frontline of patient care, play a crucial role in infection prevention and control (IPC). However, studies from Pakistan consistently report a gap between knowledge and practice of IPC measures.

Objective: To assess the knowledge, practices, and institutional barriers related to infection prevention and control among nurses working at Liaquat University of Medical & Health Sciences (LUMHS), Jamshoro and Hyderabad.

Methods: A descriptive cross-sectional study was conducted among 103 nurses at LUMHS using a structured questionnaire with 15 items covering knowledge, practice, and barriers. Data were analyzed using IBM SPSS Version 23 with descriptive statistics and chi-square tests.

Results: The majority of participants were female (61.2%), aged 26-30 years (46.6%), and held BSN qualifications (65.0%). The mean knowledge score was 3.61/5 (SD=1.02) with 60.2% demonstrating good knowledge. The mean practice score was 7.31/9 (SD=1.28) with 73.8% reporting good practice. Critical gaps included: 50.5% incorrectly believing clean water alone destroys microorganisms, 30.1% handling body fluids without gloves, and 28.2% not attending annual IPC training. No significant gender or age associations were found ($p>0.05$).

Conclusion: Nurses demonstrated adequate overall knowledge but inconsistent practice implementation. The primary barriers are institutional rather than individual, suggesting that systemic interventions targeting training frequency, guideline accessibility, and PPE supply would be more effective than knowledge-based education alone.

Keywords: Infection control, nurses, healthcare-associated infections, hand hygiene, personal protective equipment, nosocomial infections, tertiary hospital, Pakistan

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1. INTRODUCTION

Hospital-acquired infections (HAIs) remain a significant challenge in healthcare systems worldwide. Patients sometimes contract new infections during treatment, leading to worse outcomes, prolonged hospital stays, and elevated healthcare costs. These infections affect both patients and healthcare workers, particularly nurses who maintain the closest and most continuous patient contact.[1] In tertiary care hospitals handling complex cases, the presence of immunocompromised patients, invasive devices, and high patient turnover create environments where infectious agents spread more easily.[2]

Standard infection control measures address these risks effectively when properly implemented. Hand hygiene remains the cornerstone of prevention, recognized by the World Health Organization (WHO) as the single most effective measure to prevent HAIs.[3] Additional measures include proper use of personal protective equipment (PPE), safe needle and sharps disposal, regular environmental cleaning, and equipment sterilization.[4] International health organizations have published comprehensive guidelines for these practices. Yet compliance among healthcare workers, particularly in resource-limited settings, often falls short of recommended standards.[5]

Studies consistently document a persistent gap between theoretical knowledge and actual practice of infection control measures.[1,4,6] Nurses might correctly answer questions about protocols while admitting they skip proper technique during busy shifts. Several factors contribute to this gap, including lack of continuous education, inadequate training, insufficient resources, and excessive workload.[7,8]

Pakistan's healthcare system faces pressures that compound compliance challenges. Public hospitals operate with nurse-to-patient ratios below recommended levels. Supply issues leave wards without adequate protective equipment. Administrative oversight remains inconsistent.[2,5] A nurse working a 12-hour shift with twice the recommended patient load, facing glove shortages, and receiving no supervisory feedback exists in a markedly different context than the controlled settings where international protocols were developed. Tertiary hospitals treating the sickest patients represent settings where prevention failures carry the highest stakes, yet research examining these facilities in Pakistan remains limited.

1.1 Gaps in Current Research

Existing studies from Pakistan document the knowledge-practice disconnect, but important questions remain unanswered. Most research measures knowledge and practice separately without examining what prevents nurses from acting on their knowledge.

[1,2,3,4,5] Survey methods dominate, creating potential for social desirability bias. Few studies employ observational methods or examine actual compliance through direct monitoring.

The specific context of tertiary hospitals has received insufficient attention. These facilities handle referral cases, treat complex conditions, and maintain intensive care units where infection risks are elevated. Whether barriers differ between tertiary and district hospitals remains unclear. Most importantly, studies treat structural constraints as background factors rather than central inquiry subjects. When international guidelines assume consistent access to running water and unlimited sterile gloves, how do nurses in resource-limited settings prioritize competing demands?

1.2 Study Objectives

This research examines infection control among nurses in tertiary care settings through four objectives:

1. Assess current knowledge levels regarding infection prevention protocols
2. Evaluate reported infection control practices during patient care
3. Identify institutional, environmental, and individual barriers impeding adherence
4. Analyze relationships between demographics, knowledge, practices, and barriers

2. REVIEW OF RELATED LITERATURE

Khowaja and colleagues[1] conducted a cross-sectional study at Liaquat Institute of Medical & Health Sciences Hospital, Thatta, with 30 nurses. The study found that nurses demonstrated satisfactory understanding of infection control principles, but actual compliance fell below acceptable standards. Inadequate continuing education and irregular protective equipment availability were identified as key obstacles. However, the study did not explore how these barriers interact: whether guaranteed glove supply alone would improve compliance, or whether multiple factors need simultaneous change.

Batool and colleagues[2] studied 138 nurses at Sandeman Provincial Civil Hospital and Bolan Medical Complex Hospital in Quetta. They documented concerning practices including insufficient hand hygiene and found that 99.3% of participants reported using gloves and 97.8% used masks. As per perceptions, 66.7% agreed that nurses with common colds should be excused from duty, and 76.8% acknowledged that cultural attitudes influence infection spread. Environmental factors emerged as significant contributors, particularly poor sanitation and absent monitoring. Infection control practices were statistically non-significant according to age and work experience ($p>0.05$).

Kousar's team[3] studied nurses at Aziz Bhatti Shaheed Hospital, Gujrat, reporting high awareness but inconsistent implementation. Utilizing the Health Belief Model as a theoretical framework, their analysis emphasized the role of infection control committees and surveillance programs. The chi-square analysis revealed significant gender differences in knowledge, with female nurses showing higher knowledge regarding antiseptic use, autoclaving, and hand washing practices. However, the study did not examine whether institutional oversight systems actually function in Pakistani hospitals.

Soomro and colleagues[4] examined Isra University Hospital, Hyderabad, with 60 nurses, focusing on standard infection control precautions. They found high adherence to basic measures (85% for hand hygiene and standard precautions) but lower compliance with specialized procedures-only 56.7% wore gowns and gloves when contact precautions were required, and only 45% consistently wore gloves during blood collection. These findings highlight that while general preventive measures are maintained, specific high-risk procedures require additional attention.

Baqar and colleagues[5] documented occupational hazards from blood exposure among 111 nursing staff at a tertiary care hospital in Khairpur. Using a WHO-validated questionnaire combined with direct observation, they found 88.3% agreed healthcare workers should be vaccinated, but 66.7% of nurses recapped needles despite knowing the associated dangers. The study uniquely incorporated observation alongside self-reporting, revealing that 0.9%

compliance with hand washing before aseptic tasks and 19.8% after glove removal-dramatically lower than self-reported rates. Gender differences were significant (p=0.000), with female nurses showing higher compliance with hand hygiene and PPE use.

2.1 Synthesis of Evidence

Common themes emerge across these five studies. Knowledge levels appear generally adequate-most nurses can identify proper procedures when asked. Yet practice consistently falls short of knowledge, a pattern that holds across different cities and hospital types. The explanations converge on resource constraints, workload pressures, and training gaps. However, current research describes barriers without deeply analyzing their relative importance or interaction effects. Moreover, most studies rely on self-reported data, which likely inflates compliance estimates, as demonstrated by the dramatic discrepancy between self-reported and observed compliance in the Baqar study.[5]

The present study addresses these limitations by examining knowledge, practices, and barriers together within the tertiary hospital context at LUMHS, using a comprehensive instrument that captures institutional barriers alongside individual knowledge and practice measures.

3. MATERIALS AND METHODS

3.1 Study Design

A descriptive cross-sectional study was conducted to assess the knowledge, practices, and challenges of infection prevention and control among nurses working at tertiary healthcare settings.

3.2 Study Setting and Duration

The study was carried out at Liaquat University of Medical & Health Sciences (LUMHS), including the main campus hospital in Jamshoro and its branch facility in Hyderabad, Sindh, Pakistan. Data collection was conducted from January to February 2026. LUMHS is a major public sector tertiary care teaching hospital providing specialized medical services across multiple departments including Medicine, Surgery, Pediatrics, ICU, CCU, Emergency, Orthopedics, Cardiology, Pulmonology, Urology, Nephrology, Gynecology, and Operation Theater.

3.3 Sample Size Calculation

The sample size was calculated using Slovin's formula:

$$n = N / (1 + N \times e^2)$$

Where N = total nursing population at LUMHS (~140 nurses across both campuses) and e = margin of error (0.05). This yielded n = 140 / (1 + 140 × 0.0025) = 140 / 1.35 ≈ **103 participants**.

3.4 Sampling Method

Convenience (non-probability) sampling method was adopted. Nurses available and willing to participate during the data collection period were enrolled from various clinical departments across both campuses.

3.5 Inclusion and Exclusion Criteria

Inclusion: All registered staff nurses (male and female) currently employed at LUMHS in direct patient care roles. **Exclusion:** Student nurses, interns, nurses on leave during data collection, those who declined participation, and administrative nursing staff not involved in direct patient care.

3.6 Data Collection Instrument

Data were collected using a structured, self-administered questionnaire designed based on previous published literature and WHO infection control guidelines.[5,9] The questionnaire consisted of two sections:

Section A - Demographic Data (5 items): Gender, age, marital status, professional qualification, and department.

Section B - Knowledge, Practice, and Barriers (15 items): Questions were categorized into three overlapping domains: Knowledge (5 items covering HAI transmission, nosocomial infection definition, WHO hand hygiene moments, universal precautions, and water disinfection); Practice (9 items covering training availability, hand washing, PPE use, body fluid handling, and waste management); and Barriers (5 items covering training frequency, guideline accessibility, knowledge adequacy, PPE availability, and workshop attendance). Response options were Yes/No/Don't Know for all items.

3.7 Data Collection Procedure

The questionnaire was physically distributed to nurses during their duty shifts across different departments at both LUMHS campuses. The purpose of the study was explained to each participant. Participants were given adequate time to complete the questionnaire and assistance was provided for ambiguous items. Completed questionnaires were collected on the same day.

3.8 Ethical Considerations

Verbal and written informed consent was obtained from all participants. Participation was voluntary, and nurses could withdraw at any time without consequence. All responses were anonymous and confidential. The study adhered to the principles of the Declaration of Helsinki. Approval was obtained from hospital administration.

3.9 Data Analysis

Data were entered and analyzed using IBM SPSS Statistics Version 23. Descriptive statistics (frequencies, percentages, mean, standard deviation) were used to summarize demographic characteristics and response distributions. Chi-square test (χ^2) was used to test associations between demographic variables and knowledge/practice scores. Cross-tabulation was used to compare responses across demographics. A p-value ≤ 0.05 was considered statistically significant.

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